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SUITMA 8: SOILS AND SEDIMENTS IN URBAN AND MINING AREAS

# Ammonium-nitrate dynamics in the critical zone during single irrigation events with untreated sewage effluents

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**Abstract**
*Purpose* Previous studies in the Mezquital Valley evidenced that irrigation with untreated sewage effluent supplies two- to tenfold larger nitrogen doses to crops than common fertilizer recommendations. However, nitrate concentrations in the groundwater are only slightly above threshold concentrations for drinking water. To understand the N dynamics in this agroecosystem, we quantified nitrogen inputs, outputs, and transformations within the rooting zone and in the vadose zone down to the aquifer (i.e., in the critical zone).
*Materials and methods* Single irrigation events were monitored in three different fields cropped with either annual rye grass (*Lolium rigidum*) or oats (*Avena sativa* L.) harvested for fodder. For each irrigation event, the total amount of water entering and leaving the field was quantified with a flowmeter. Soil pore water was collected with either microsuction cups or observation wells and groundwater was sampled at two wells. All water samples were analyzed for total nitrogen (Nt), ammonium nitrogen ($NH_4^+$–N), nitrate nitrogen ($NO_3^-$–N), chloride ($Cl^{-1}$), and pH. Organic N was calculated as the difference between total N and inorganic N. The water tension and the soil water content were monitored before, during, and after the irrigation with tensiometers and TDR probes, respectively, installed at different depths and at three sites within each field. Batch experiments were conducted to assess the $NH_4^+$ adsorption capacity of the soils.
*Results and discussion* The irrigations added 537 to 727 kg ha$^{-1}$ N in form of organic N (40 %) and $NH_4^+$–N (60 %) to the fields. Crops absorbed 65 % of the N and 31 to 66 kg $NO_3^-$–N ha$^{-1}$ leached out beyond the rooting zone (>40 to 130 cm). Batch experiments evidenced an ammonium adsorption capacity of 43 and 53 % of the input ammonium mass. Nitrification dominated over denitrification as the water infiltrated through the soil, evidenced by changes in nitrate concentrations and pH values in the soil pore water. The behavior of the total N/Cl ratio with depth indicated possible N losses due to $NH_3$ volatilization at the field surface, a temporal withdrawal of N from the soil solution due to $NH_4^+$–N adsorption in the rooting zone, as well as probable denitrification losses in the vadose zone.
*Conclusions* Although the studied agroecosystem muses the large N inputs relative efficiently, between 7 and 10 % of the added N with each irrigation leaches beyond the crop root zone as nitrate. This is triggered by overflow irrigation, since up to 8,699,000 L of water with N concentrations of up to 50 mg total N L$^{-1}$ infiltrate rapidly through macropores beyond the rooting zone. Additionally, ammonia volatilization and denitrification seem to be occurring. The latter could provide a self-cleaning potential to the system, if it reaches $N_2$ and needs further verification. Nevertheless, N inputs to the system should match crop uptake to avoid groundwater and atmospheric pollution.

**Keywords** Ammonia volatilization · Ammonium adsorption · Denitrification · Groundwater pollution · Nitrate leaching

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## 1 Introduction

Nitrogen is an essential plant nutrient and is therefore added in the form of fertilizer or organic waste amendments to agricultural fields to improve crop yields. However, N application in excess is detrimental to grain yield, since it enhances vegetative growth and thus lodging and retards grain maturation (FAO 1985). It also promotes the appearance of pests and diseases in cropped fields (Kass 1996; Reid et al. 2001). N surplus also causes environmental damage in several ways: it is susceptible to be transferred into the atmosphere by either ammonia volatilization or denitrification, if the soil pH is alkaline or soil aeration is deficient, respectively. Ammonia is a precursor of acid deposition and also known to contribute to ecosystem eutrophication (Skeffington and Wilson 1988; Camargo and Alonso 2006). Denitrification is harmless if it reaches the air as elemental $N_2$. However, the denitrification process can be incomplete so that the greenhouse gas $N_2O$ is emitted into the atmosphere (EPA 2015). This gas contributes 250- to 310-fold more to global warming than $CO_2$, and over-fertilization of agricultural crops has been recognized as its primary source (Mei et al. 2004; Pihlatie et al. 2004). Excess N can also be leached out of the rooting zone in the form of nitrate and pollute the groundwater (Withers and Lord 2002; Shomar et al. 2008; Dungait et al. 2012; Ferguson 2015). Ingestion of water contaminated with nitrate is known to cause methemoglobinemia, a disorder that decreases the blood's ability to bind oxygen, and affects particularly infants younger than 6 months (Karr 2012). These do not have yet the capacity to produce an enzyme that reduces methemoglobin in their blood, and are therefore especially at risk if they consume polluted drinking water. But also, adults are at risk of acquiring stomach and bladder cancer if ingested nitrates are transformed into nitrosamines in their gastrointestinal tract (Du et al. 2007; Ward et al. 2005; Gupta et al. 2008). To prevent these health problems, many countries and the World Health Organization (WHO) have established maximum permissible limits of $NO_3^-$–N concentrations in the groundwater of 11.3 mg $L^{-1}$ (WHO 2008).

Agriculture is widely acknowledged as the major nitrate source to groundwater (Diez et al. 1997; Costa et al. 2002). Main sources are nitrogen fertilizers and manure, particularly liquid swine and cattle manure, but also wastewater irrigation with either untreated or treated sewage effluents. Any of these N sources has the potential to pollute air and groundwater if the N load applied to the field surpasses the N uptake by the crop. A risk assessment of N losses from an agroecosystem should additionally consider the timing of the N application along the crop cycle. In sewage effluent irrigated systems, N is applied with each irrigation and not considering the growth stage of the crop. Annual crops as maize (*Zea mays*) need several weeks after sawing until they develop enough root biomass to absorb the applied N and are much more susceptible to contribute to N losses than perennial fodder crops. Also, the form in which the fertilizer is applied matters: more soluble inorganic salts or easily mineralizable organic amendments will release N at faster rates than crops can take up. In sewage effluents, N is present mostly as $NH_4$–N or in the form of easily mineralizable organic compounds with a C/N ratio smaller than 4 (Feigin et al. 1991). Sewage effluents are irrigated mostly by overflow, particularly if they have large contents of suspended matter, which clogs sprinklers or drip irrigation devices. However, overflow irrigation promotes rapid infiltration of large water volumes along cracks at the beginning of an irrigation event, moving eventually water and nutrients beyond the rooting zone. At the end of an irrigation event, the soil will be saturated with water, so that denitrification can take place, which is triggered additionally by the presence of easily mineralizable organic matter (González-Méndez et al. 2015).

Site properties also play a role in determining the fate of N from fertilizer and organic amendments in agroecosystems. The risk of nitrate leaching is particularly large in coarse-textured soil (Di and Cameron 2002; Gurdak et al. 2007), in areas with shallow aquifers, and locations prone to receive frequent and intense rain or irrigation events during the first stages of crop development (Chesnaux and Allen 2008; Wang et al. 2010). Particularly irrigation by overflow, where large amounts of water percolate in short time through the soil, triggers aquifer enrichment with nitrate (Hamilton et al. 2007; Jiménez and Asano 2008; Raschid-Sally and Jayakody 2008). Concentrations as large as 388 mg $L^{-1}$ have been reported in the infiltration water beyond the rooting zone of a silty clay loam soil irrigated with treated sewage effluent sand cultivated with sugar cane (Pereira Leal et al. 2010). However, in general, there are only few investigations dealing with the nitrogen dynamics in agroecosystems that use untreated sewage effluents for irrigation, in comparison with abundant studies reporting nitrate leaching from agroecosystems amended with animal manure (Sophocleous et al. 2010; Rahil and Antonopoulos 2007), and their results are contradictory.

The reuse of either treated or untreated wastewater, for irrigation, is increasing worldwide to optimize water use and to recycle nutrients. The fact that large amounts of water are applied together with the nutrients to the fields leads to an expectation of important nitrate loads to aquifers by these so-called soil aquifer treatment systems (Wlodarczyk and Kotowska 2006; Asano et al. 2007; Stenger et al. 2008; Yaron et al. 2008; Derby et al. 2009; Keesstra et al. 2012).

On the other hand, it is well known that soil properties such as the pH, the overall biological activity, the soil organic matter content, the soil texture and its relation with the water retention capacity, the aeration capacity, and the adsorption capacity of ammonium are determinants of the N dynamics within the soil. Such processes as $NH_3$ volatilization, N oxidation to nitrate or reduction to nitrous oxide and elemental N, and N retention and adsorption depend on these properties (Gouveia and Eudoxie 2002; Abel et al. 2012). Also, other site conditions as precipitation, evaporation, and land management influence N dynamics. Nitrate leaching out of the rooting zone does not necessarily reach the groundwater, since on its way through the vadose zone, still nitrate reduction to elemental $N_2$ and also consumption by microbes can take place (Garg et al. 2005; Chesnaux and Allen 2008). The latter provides a self-cleaning capacity that needs to be accounted for when assessing nitrate leaching hazards.

The objectives of this investigation were (a) to quantify by field measurements N inputs (irrigation water), outputs (crops uptake, surface and subsurface lateral outflows of the field, and vertical outflows beyond the rooting zone), and N transformations in an agroecosystem that has been using untreated wastewater for irrigation for more than 80 years in Central Mexico; (b) to determine the ammonium adsorption capacity of the soil through batch experiments in the laboratory; and (c) to assess the N use efficiency, the nitrate leaching potential to the aquifer, and the self-cleaning potential of the system. We hypothesized that in land use systems irrigated with untreated sewage effluents N dynamics are determined by the large amount of deep percolating water during and shortly after irrigation, the changes in N speciation from $NH_4^+$–N in the irrigation water to $NO_3^-$ within the aerated soil and to nitrous oxide and $N_2$ in the saturated soil (Fig. 1). The latter together with the $NH_4^+$–N adsorption capacity of medium- to fine-textured soils could provide an important self-cleaning capacity to this system. Since most effluents have alkaline pH values, we also expect some ammonia volatilization as the water flows over the field (Fig. 1).

To test these hypotheses, we monitored single irrigation events in three different fields and monitored water inflows and outflows, soil water tension, and water content at different depths, as well as concentrations of N species in the soil solution, the subsurface lateral outflow, and in soil pore water within the vadose zone down to the groundwater. In the water samples, we also measured pH and in the groundwater dissolved organic carbon (DOC) concentrations, as determinants and indicators of ammonia volatilization, nitrification, and denitrification. To account for mean gains or losses of nitrogen along the water flow path, we calculated the ratio between total N and $Cl^-$ as conservative tracer. This was particularly of use, since losses in form of $N_2$ are otherwise not identifiable under field conditions. With batch experiments, we determined the soils $NH_4^+$–N adsorption capacity, and by quantifying the aboveground biomass and its N contents, we assessed the N use efficiency.

## 2 Materials and methods

### 2.1 Description of the land use system

The Mezquital Valley, in the southeast of the state of Hidalgo, is the largest continuous area irrigated with untreated sewage effluents in the world (Almanza-Garza 2000; Raschid-Sally and Jayakody 2008). Irrigation started more than a century ago and is pursued with the sewage effluents and surface runoff of the metropolitan area of Mexico City (BGS 1998; Jiménez et al. 2005). The irrigated area has been growing as the discharge volumes provided by the increasing population in the basin of Mexico also grow. Today, 40 $m^3$/s of untreated sewage mixed with 12 $m^3$/s of surface runoff is discharged to the valley and more than 90,000 ha are irrigated (BGS 1998). The region has a semiarid climate with a mean annual precipitation of 459.8 mm, a mean annual temperature of 17.3 °C, and a mean annual evaporation of 1896.9 mm (CNA 2015). Dominant crops are alfalfa (*Medicago sativa* L.) and maize (*Z. mays* L.) cultivated in 3 and 2 years rotation, maize being followed by mainly rye grass (*Lolium rigidum*) or oats (*Avena sativa* L.) harvested for fodder as second crop. Alfalfa receives between 10 and 12 irrigations per year, maize receives 6, and oats 4, while rye grass is irrigated every 15 days during 3 months. Each irrigation varies between 15 and 25 mm, depending on soil moisture conditions.

Nitrogen concentrations in the irrigation water range between 30 and 50 mg $L^{-1}$, and the dominant N species is $NH_4^+$–N followed by organic N (SARH 1985; Chávez et al. 2012; and own unpublished data). The mean yearly N inputs with the irrigation are of 527 and 326 kg $ha^{-1}$ to alfalfa (*M. sativa* L.) and maize (*Z. mays* L.), respectively (Siebe 1998). These N inputs clearly exceed the needs of alfalfa, since this crop has the ability to fix atmospheric nitrogen through its symbiosis with bacteria (*Sinorhizobium* sp.) and only needs eventually a N fertilization of 40 kg $ha^{-1}$ at the beginning of the crop cycle, if irrigated with well water. Fertilization recommendations to rain-fed maize by local institutions are of 180 kg N $ha^{-1}$ (Siebe 1998), those of oats of 250 kg N $ha^{-1}$, and those of rye grass 150 kg N $ha^{-1}$ (references). Irrigation has increased 25-fold the natural aquifer recharge in the region (BGS 1998), which is of 165 $Mm^3$ $year^{-1}$, of which only 15.8 $Mm^3$ $year^{-1}$ are provided by precipitation, while 69 $Mm^3$ $year^{-1}$ correspond to deep percolation of irrigation water and 80.2 $Mm^3$ $year^{-1}$ to leakage from unlined irrigation canals (Lesser-Carrillo et al. 2011). An unconfined shallow aquifer has formed within the Tarango formation, which is the youngest volcaniclastic deposit in the valley after the alluvium that provides the soil's parent material (Del Arenal-Capetillo 1985; BGS 1998).

Reported nitrate concentrations in the groundwater range between 6.1 and 29 mg $L^{-1}$ (Jiménez and Chávez 2004). The BGS (1998) reported that in 25 % of 50 evaluated wells, the

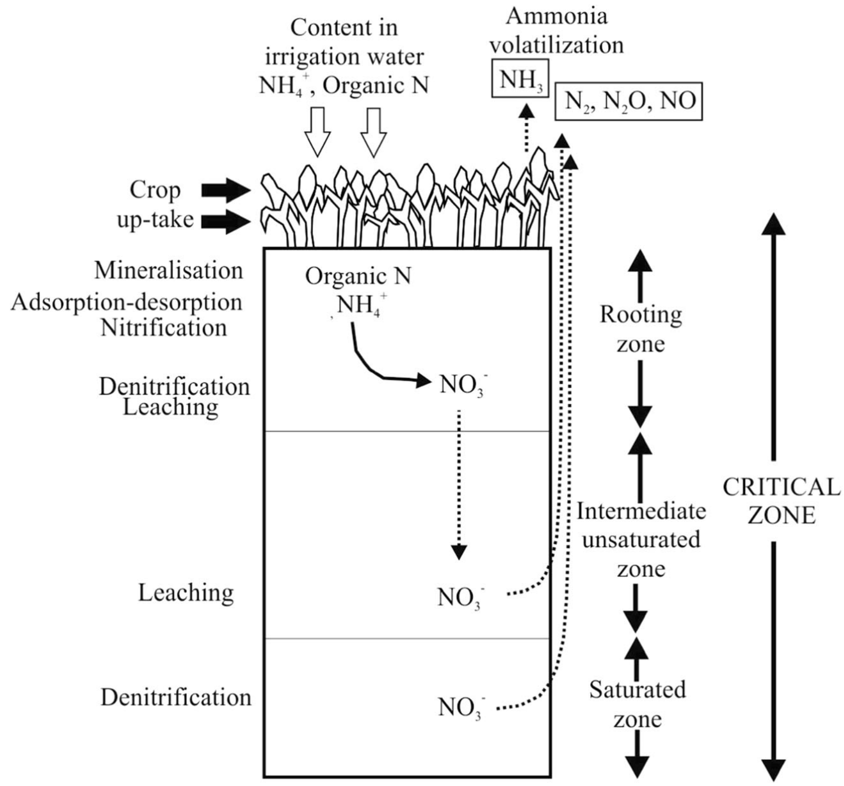



**Fig. 1** Conceptual model of N dynamics in the studied agroecosystem

maximum permissible limit of 11.3 mg $L^{-1}$ $NO_3^-$–N established by the WHO for drinking water was exceeded; concentrations ranged between 11.6 and 15.5 mg $L^{-1}$ of $NO_3^-$–N. Currently, the groundwater is used as water supply for 500,000 people living within the valley.

### 2.2 Studied fields

The studied fields are located on an extended volcanic piedmont near Tlahuelilpan. The altitude within the piedmont ranges between 2070 and 2090 masl (Fig. 1), and the slope is moderate, 2° to 5° (°), so that fields have been terraced in order to achieve a slope of <2°.

The soils classify as haplic or vertic Phaeozems, which developed on colluvio-alluvial sediments deposited on top of a series of layered volcaniclastic tuff deposits of Tertiary age, locally known as Tarango Formation. These materials have a medium to low permeability, and their thickness varies between 0.5 and 7 m within the piedmont (Hernández-Martínez et al. 2014). The groundwater flow is from east to west, and the groundwater level is at 22 to 25 m below the surface (BGS 1998). At 120 and 1000 m distance of the selected fields, two open wells are located, so-called norias, at which the groundwater was sampled at a depth of 32 and 22 m, respectively (N1 and N2 in Fig. 2). Irrigation water is provided to the fields by the Tlamaco-Juandhó channel (Fig. 2), which flows south–north at 2090 m. Irrigation is carried out by field overflow.

Three fields were chosen in the middle part of the piedmont (Fig. 2), and one irrigation event was monitored in each of them in the month of January of the years 2012, 2013, and 2014. The size of the fields was 2.5, 2.0, and 2.6 ha, respectively. In 2012, the chosen field was cultivated with fodder oats (*A. sativa* L.), and in 2013 and 2014, both fields were cultivated with rye grass (*L. rigidum*). Fodder oats (*A. sativa* L.) received four irrigations (one each month), and they were harvested after 4 months. Rye grass (*L. rigidum*) was irrigated every 15 days; it was first harvested after 3 months, and the second and third cut was done 4 and 8 weeks afterwards, respectively. Neither fertilizer nor pesticides were added to both crops.

In 2012, three soil pits were excavated within the field, one near to the water inlet, one in the middle of the field, and one near to the water outflow of the field (Fig. 2). In 2013 and 2014, only two pits were excavated at the water inlets and outlets of the fields, respectively. Soil profiles were described (Siebe et al. 1996) and sampled by genetic horizon and analyzed in the laboratory using standard procedures (Van Reeuwijk 1992). Field descriptions provided information about the rooting depth, which varied between 40 cm in profile 1 of the field monitored in 2014 to a maximum of 130 cm in profile 2 of the field monitored in 2012.

The yield of fodder oats (*A. sativa* L.) was quantified by collecting the aboveground biomass few days before harvest at six spots of 0.5 $m^2$ chosen at random within each field. The collected biomass was weighted, dried at 60 °C, and weighted

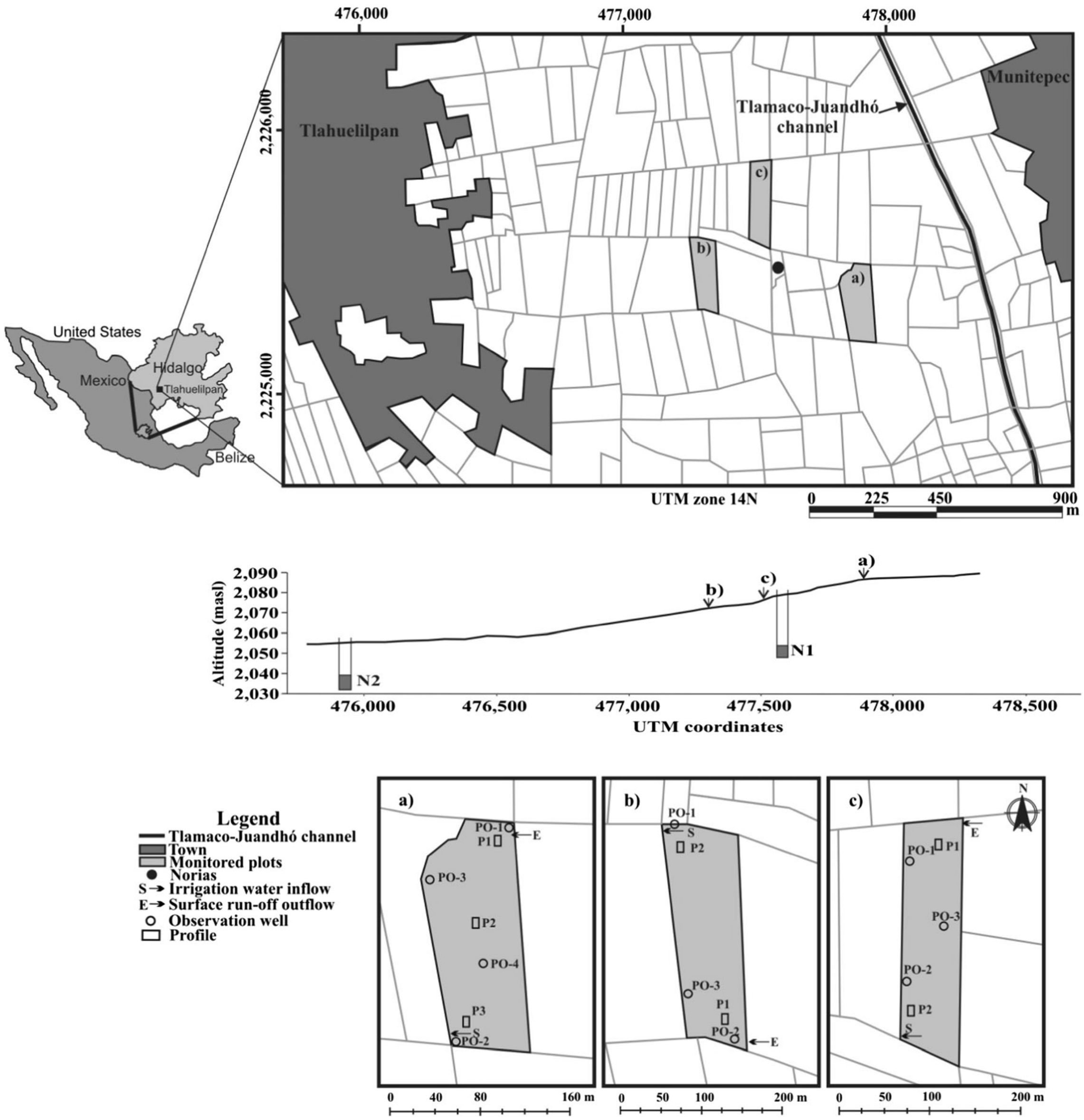



**Fig. 2** Dimensions of the studied fields and their location within the piedmont. The figure shows also the location of the soil profiles within each field, as well as the location of the open wells and the depth to the ground water table. **a** Plot monitored in 2012, **b** plot monitored in 2013, and **c** plot monitored in 2014

again to determine fresh and dry mass, and an aliquot was ground for N analysis (CNHS/O PerkinElmer 2400 series II).

### 2.3 Field measurements

In the soil, we monitored the soil water content and its tension before, during, and after each irrigation. For this, we installed tensiometers at 10–15, 30–35, and 55 cm depth close (30 cm) to each soil pit. Soil water content was measured at the same sites and depths with TDR probes (TRASE TDR, Soil Moisture Equipment Corporation). Additionally, we installed three microsuction cups (MacroRhizon$^{MR}$) at 10, 20, and 30 cm depth in each profile, from which we sampled the soil pore water at time intervals between 1 and 2 h during irrigation and after irrigation each 4 h. The soil pit located in the middle of the field monitored in 2012 was kept open during the irrigation; to avoid irrigation water to flow into the pit, we installed metallic sheets of 30 cm height all around the pit to divert drainage water. This allowed us to collect soil pore water at 30 cm depth draining from soil macropores during the irrigation (i.e., preferential flow water).

During and after each irrigation, the subsurface lateral flow was quantified and sampled in three observation wells. One was installed in the middle upper part of each field and the other two close to both lower corners of each field (Fig. 2). These observation wells were installed at the following depths: 90–120, 200–230, near 300, and down to 400 cm, below a compacted layer identified either by the appearance of hydromorphic properties or a higher moisture content. These wells were drilled with Eijelkamp drills until the desired depth, then a PVC tube of 5 cm diameter, which was slotted in its lowest 50–100 cm, was introduced into the hole and the space between the hole and the slotted tubing was filled with washed quartz sand and sealed with bentonite above the slotted portion of the tubing up to the soil surface to prevent irrigation water intrusion along the tubing.

The amount of water entering and leaving the field by surface runoff was estimated by measuring the flux with a fluxmeter (Global Water). Subsurface lateral flow leaving the field was determined by measuring of the height of the water table in the observation wells at periodic time intervals during and after the irrigation event until the flow stopped (VenTe et al. 2000).

The water volume retained in the soil was estimated by multiplying the soil moisture content at the different depths in each soil profile 24 h after each irrigation event considering the thickness of the horizon in centimeter and by adding all depths. We measured the solum depth within the field with an auger at 18 locations placed at regular grid intervals over the whole field. This information was used to assign the area represented by each profile and to extrapolate water volume measured near each soil pit to the whole field (Table 1). The amount of water that left the field by deep percolation was estimated by subtracting surface outflow, water retained in the soil, and the subsurface lateral flow from the total volume entering the field.

### 2.4 Water sampling

Three irrigation water samples were collected during each event at the water inlet to the field. Also three samples of the surface outflow of the field were collected during each irrigation. Subsurface flow was sampled at the observation wells by pumping out the water during irrigation and up to 30 to 48 h after it, as long as there was water inside the observation wells. Groundwater samples were collected from two deep wells also by pumping. All samples were immediately filtered through 0.45-μm (Millipore) membrane filters, stored at 4 °C, and transported to the laboratory for the analysis of major ions. In the field, we determined the pH, oxidation-reduction potential, electric conductivity, and the temperature with a portable multiparametric device (Hanna Instruments 9218). Aliquots of the groundwater samples were immediately acidified with 6 N HCl in the field and transported also at 4 °C for the analysis of DOC.

### 2.5 Laboratory analyses

Particle size distribution of the soil samples was determined by the combined sieve and pipette method after eliminating the organic matter with $H_2O_2$ treatment (Schllichting et al. 1995). The soil pH was measured in soil water suspensions 1:2.5 (wt/vol) in distilled water with a Beckman 34 potentiometer (ISRIC 1992). The total contents of C and N were determined with an elemental analyzer (CNHS/O PerkinElmer 2400 series II), calibrating with the standard reference material LECO CNS 2000.

The anions ($HCO_3^-$, $Cl^-$, $NO_3^-$, $SO_4^{2-}$, $F^-$) and cations ($Na^+$, $NH_4^+$, $K^+$, $Ca^{2+}$, $Mg^{2+}$) in water samples were determined by ion chromatography in a Waters 1525 chromatograph equipped with a binary pump, an autosampler (717), and an electric conductivity detector (432). The stationary phase used for the anion analyses was a IC-PaK column of 4.6 × 75 mm (Waters), and the mobile phase was acetonitrile/butanol/gluconateborate solution of sodium/water (12:2:2:84) in isocratic mode and with a flux of 1 ml min$^{-1}$. Cations were quantified using as stationary phase a Metrosep C4 column (Metrohm) of 4 × 100 mm, and the mobile phase was a 1.9 mM $HNO_3$ solution with 0.8 mM dipicolinic acid in isocratic mode and a flux of 0.9 ml min$^{-1}$. The total N concentrations in all samples and the DOC contents in groundwater samples were determined with an elemental analyzer (Shimadzu TOC-L analyzer).

Batch experiments to determine the $NH_4^+$–N adsorption capacity were conducted with samples of the upper 30 cm of the soils sampled in each plot (Roy et al. 1991). Duplicate samples of 1.5 g air-dried soil were mixed with 30 ml of a background electrolyte solution and spiked with ammonium in form of $NH_4$-acetate ($CH_3COONH_4$) in the following concentrations: 3.9, 11.7, 23.3, 31.1, 62.2, and 93.4 mg L$^{-1}$ of N–$NH_4^+$, which were chosen beyond and below the measured

**Table 1** Thickness of the solum in the monitored profiles and percentage of the total field area represented by each of the profiles

| Year | Profile 1 (P1) | | Profile 2 (P2) | | Profile 3 (P3) | |
|---|---|---|---|---|---|---|
| | Thickness (cm) | Percentage of field area (%) | Thickness (cm) | Percentage of field area (%) | Thickness (cm) | Percentage of field area (%) |
| 2012 | 72 | 50 | 124 | 30 | 106 | 20 |
| 2013 | 130 | 35 | 80 | 65 | ND | ND |
| 2014 | 40 | 25 | 93 | 75 | ND | ND |

*ND* not determined

$NH_4^+$–N concentrations in the irrigation water (23.3 and 31.1 mg $L^{-1}$). The tubes were equilibrated by shaking for 24 h at 25 °C, then centrifuged at 2500 rpm for 30 min, and the $NH_4^+$–N concentration was determined in the supernatant. The background electrolyte was a reconstituted solution with the major cations and anions found in the irrigation water, namely KCl, $CaCl_2$, $MgCl_2$ and $NaHCO_3$, and $NH_4^+$–N at concentrations of 0.73, 0.78, 0.80, and 7.20 mmol $L^{-1}$, respectively.

The adsorption capacity constant ($K_f$) was determined using Eq. (1):

$$\mathrm{Log}(q) = \log(K_{\mathrm{f}}) + 1/n\,\log(Cs) \quad (1)$$

where $q$ (mg $kg^{-1}$) is the adsorbed quantity of $NH_4^+$–N and $C_s$ (mg $L^{-1}$) the concentration in solution. The percentage of $NH_4^+$–N adsorbed in each point of the isotherm was calculated with Eq. (2):

$$\mathrm{N{-}NH_4^+\ adsorbed}(\%) = [(C_{\mathrm{o}} - Cs)/C_{\mathrm{o}}] \times 100 \quad (2)$$

where $C_o$ (mg $L^{-1}$) is the initial concentration of $NH_4^+$–N.

## 3 Results and discussion

### 3.1 Soil characterization

The soils of the three monitored fields have a clayey texture in the surface horizon with more than 45 % clay (Table 2). All soils have slickensides at medium depth, and the dominant clay minerals are smectites (Siebe 1994). The bulk density in the surface horizons is low (1.0 g $cm^{-3}$) and increases with depth (to up to 1.4 g $cm^{-3}$). The C horizon has a high bulk density (>1.7 g $cm^{-3}$). The soil pH is alkaline (range, 7.16 to 8.87), organic C and total N contents in the first 30 cm are large (C, 1.95–3.20 %; N, 0.24–0.34 %), and the C/N ratio varies between 8 and 11.8. Roots could penetrate the soil without physical or chemical limitations in each soil profile through all horizons until the volcanic tuff layer of the Tarango Formation was reached. Rooting depth varied between 40 and 130 cm depth.

### 3.2 Estimated water balance for each irrigation event

The irrigation applied to the fields monitored in 2012 and 2013 was similar (0.25–0.26 m), while the irrigation of the field monitored in 2014 was smaller (0.15 m) (Table 3). Between 8 and 28 % of the applied water left the field again as surface outflow, while 49 to 79 % was retained in the soil, yielding an irrigation efficiency of 49 to 79 %. Between 0.01 and 0.15 % of the irrigation left the field by subsurface lateral flow and 13 to 22 % by deep percolation. Multiplication of these volumes with the mean number of irrigations ($n = 11$) yields an aquifer recharge of 2414 to 8593 $m^3$ $ha^{-1}$ each year. Extrapolated over the whole irrigation district, the aquifer recharge would correspond to 6.9 $m^3$ $s^{-1}$, if a mean irrigation per event of 0.15 m is assumed, and of 24.5 $m^3$ $s^{-1}$ when each irrigation event is of 0.26 m. These data show that the irrigation efficiency improves, when smaller volumes are applied per irrigation, and that deep percolation increases when more than 0.15 m of water is applied. Estimated water losses due to evapotranspiration during the

**Table 2** Physical and chemical properties of the different soil horizons of the studied profiles within each field

| Year | Depth (cm) | pH water (1:2.5) | C (%) | N (%) | Particle size distribution (%) | | | Bulk density (g $cm^{-3}$) |
|---|---|---|---|---|---|---|---|---|
| | | | | | Clay | Silt | Sand | |
| 2012 | 0–38 | 7.16 | 2.33 | 0.22 | 46.2 | 50 | 3.9 | 1 |
| | 38–60 | 7.94 | 0.94 | 0.09 | 40.4 | 54.3 | 5.3 | 1 |
| | 60–82 | 8.02 | 0.59 | 0.05 | 32 | 48.5 | 19.6 | 1.3 |
| | 82–98 | 7.84 | 0.8 | 0.07 | 29.6 | 50.9 | 19.5 | 1.4 |
| | 98–106 | 7.7 | 0.66 | 0.07 | 25.4 | 49.1 | 25.5 | 1.3 |
| 2013 | 0/20–17/25 | 7.24 | 3.2 | 0.34 | 47.4 | 45.5 | 7.2 | 1 |
| | 17/25–45/50 | 8.08 | 1.61 | 0.16 | 42 | 44.7 | 13.3 | 1.1 |
| | 45/50–90/99 | 8.13 | 1.08 | 0.11 | 38.4 | 41.6 | 20.1 | 1.2 |
| | 90/99–130 | 8.32 | 0.84 | 0.08 | 36 | 44.7 | 19.3 | 1.1 |
| | 130–165 | 8.67 | 0.4 | 0.05 | 30.4 | 47.1 | 22.5 | 1.2 |
| 2014 | 0–30 | 8.25 | 1.95 | 0.24 | 47.7 | 35.1 | 17.1 | 0.9 |
| | 30–40 | 8.62 | 0.93 | 0.11 | 49.6 | 40.5 | 10 | 1 |
| | 40–55 | 8.87 | 0.88 | 0.09 | 23.6 | 55.8 | 20.6 | 1.1 |

**Table 3** Field water balance for each irrigation event in each of the monitored fields

| | | Year | | |
|---|---|---|---|---|
| | | 2012 | 2013 | 2014 |
| Field area | (ha) | 2.5 | 2 | 2.6 |
| Volume entering the field | ($m^3$ $ha^{-1}$) | 3480 | 3271 | 1688 |
| Surface outflow | ($m^3$ $ha^{-1}$) | 982 | 666 | 139 |
| Volume retained in the field (solum[a]) | ($m^3$ $ha^{-1}$) | 1713 | 1926 | 1327 |
| Deep percolation water | ($m^3$ $ha^{-1}$) | 781 | 679 | 219 |
| Subsurface lateral flow | ($m^3$ $ha^{-1}$) | 3.2 | 0.3 | 2.5 |
| Irrigation | (m) | 0.25 | 0.26 | 0.15 |

[a] See Table 1

irrigation event (4 to 6 h) were of 0.5 to 0.8 mm (Hargreaves and Samani 1985), which does not affect the water balance significantly.

### 3.3 Nitrogen dynamics

The irrigation water is alkaline (7.78–8.00), lacks dissolved oxygen (<1 mg $L^{-1}$), and has strongly reducing conditions (Eh close to −50 mV) (Table 4); the dominant nitrogen species are $NH_4$–N (26 to 30 mg $L^{-1}$) and organic N (Nt–$NH_4^+$–N = 20–27 mg $L^{-1}$) (Table 4).

Ammonium is readily transformed into nitrate at the soil surface and as the irrigation water infiltrates into the soil and flows across the field (Fig. 3). During the first 3 h of irrigation, as the water percolates into the still aerated soil, nitrate concentration increases and the pH of the soil pore water drops by 0.2 to 0.7 units, as exemplified in the three profiles monitored in 2012 (Fig. 4a–c). However, as the soil saturates with water, which is evidenced by the falling water tension (Fig. 4d), nitrate concentrations diminish temporarily, until free water drains out of the soil macropores, causing the water tension and nitrification to increase again, particularly in the top 20 cm of the soil profile.

Nitrate concentrations reach values of up to 60 mg $L^{-1}$, and the drop in pH corresponds to the expected proton production by nitrification of ammonium under oxidizing conditions (Navarro and Navarro 2003; Tortora et al. 2007) as shown in Eq. (3):

$$2O_2 + NH_4^+ \rightarrow H_2O + NO_3^- + 2H^+ \quad (3)$$

The same behavior was observed in the fields irrigated in 2013 and 2014, respectively (Fig. 3).

In the short time interval of 3 h during which the soil has anoxic conditions, the nitrate concentrations diminish and the pH increases most probably due to denitrification according to Eq. (4):

$$2\,NO_3^- + 10\,e^- + 12\,H^+ \rightarrow N_2 + 6H_2O \quad (4)$$

González-Méndez et al. (2015) measured $N_2O$ emissions from nearby fields cultivated with alfalfa, maize, and rye grass, particularly during and 1 day after irrigation events, indicating that denitrification is incomplete. The fact that $NH_4^+$–N cannot be detected during this time interval is an additional indication that denitrification is responsible for the observed change in pH and the nitrate concentration drop.

Miller et al. (2006) observed increased nitrification after irrigation events performed by flooding. They attributed the increase in nitrification to the release of adsorbed ammonium after the irrigation event. We determined the $NH_4^+$–N adsorption capacity of the studied soils in batch experiments considering the ammonium concentration ranges measured in the influent (23.3 and 31.1 mg $L^{-1}$ of $NH_4^+$–N). These experiments were performed with soil samples collected from the fields monitored in 2012 and 2014 (Table 5). We adjusted Freundlich equations to the data, and the calculated $K_f$ values varied between 43 and 36 L $kg^{-1}$. The percentage of the adsorbed $NH_4^+$–N was 54 and 49 % of the estimated input ammonium mass, considering concentrations of 23.3 and 31.1 mg $L^{-1}$ $NH_4^+$–N in the irrigation water, respectively.

**Table 4** Characteristics of the irrigation water applied to the fields in 2012, 2013, and 2014

| | 2012 Mean and (Std. Dev) | 2013 Mean and (Std. Dev) | 2014 Mean and (Std. Dev) |
|---|---|---|---|
| pH | 8.02 (0.02) | 7.83 (0.16) | 7.78 (0.19) |
| DO, mg $L^{-1}$ | 0.6 (0.2) | 0.4 (0.1) | 0.2 (0.2) |
| Eh, mV | −55.8 (26.5) | −46.4 (29.7) | −52.2 (32.2) |
| $NH_4^+$–N, mg $L^{-1}$ | 30.8 (5.2) | 26.0 (6.1) | 26.0 (4.0) |
| $NO_3^-$–N, mg $L^{-1}$ | 0.01 (0.01) | 0 | 0 |
| Nt, mg $L^{-1}$ | 50.6 (7.6) | 53.4 (11.2) | 44.0 (4.2) |
| DOC | 49.5 (1.5) | 47 (3.0) | 50 (1.3) |
| $Cl^-$, mg $L^{-1}$ | 114 (2.3) | 126.7 (0.6) | 172.7 (17.7) |

Mean values and standard deviations (in brackets) of three samples per irrigation event

*DO* dissolved oxygen, *Eh* oxidation-reduction potential, *Nt* total N, *DOC* dissolved organic carbon

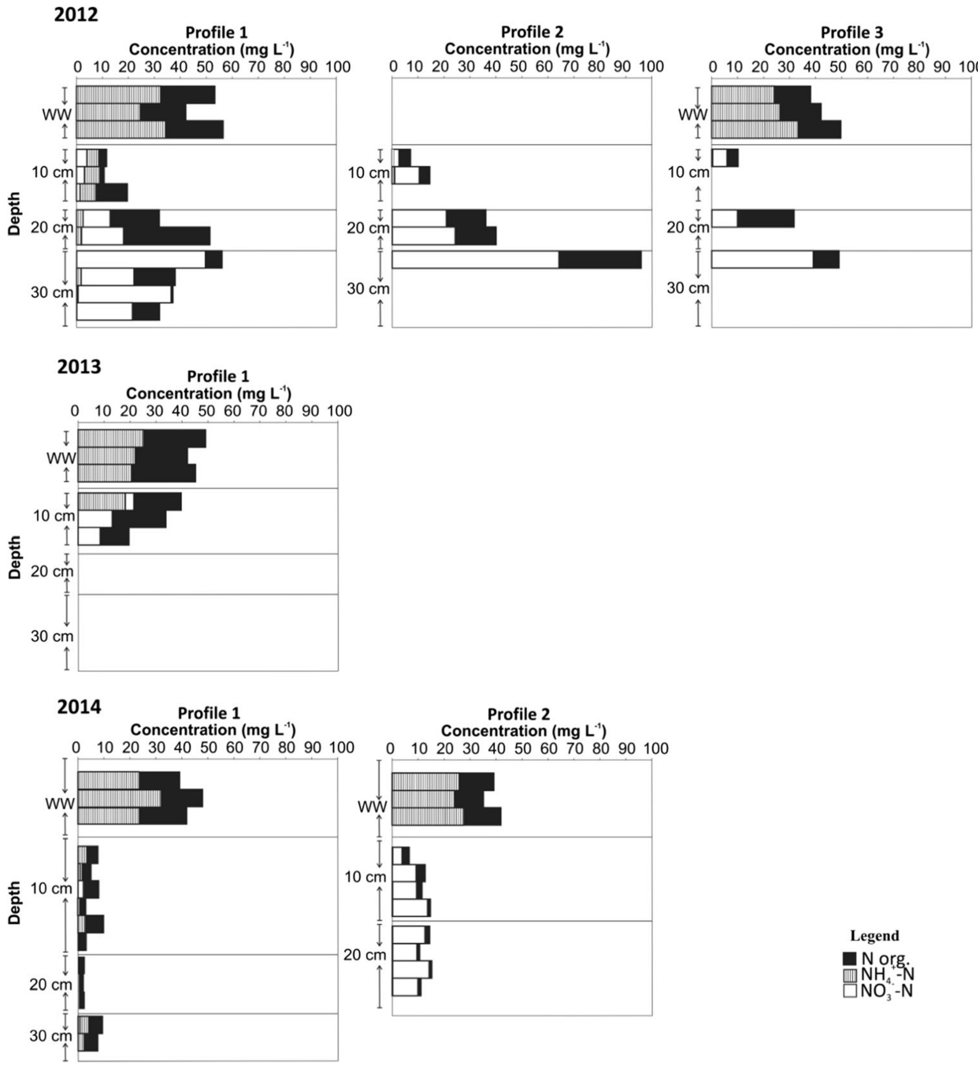



**Fig. 3** Mean concentrations of N species ($NO_3^-$–N, $NH_4^+$–N, and N org) in the irrigation water and in the soil pore water collected at different depths of the three soil profiles monitored in 2012 during the first 3 h of the irrigation event. *WW* wastewater

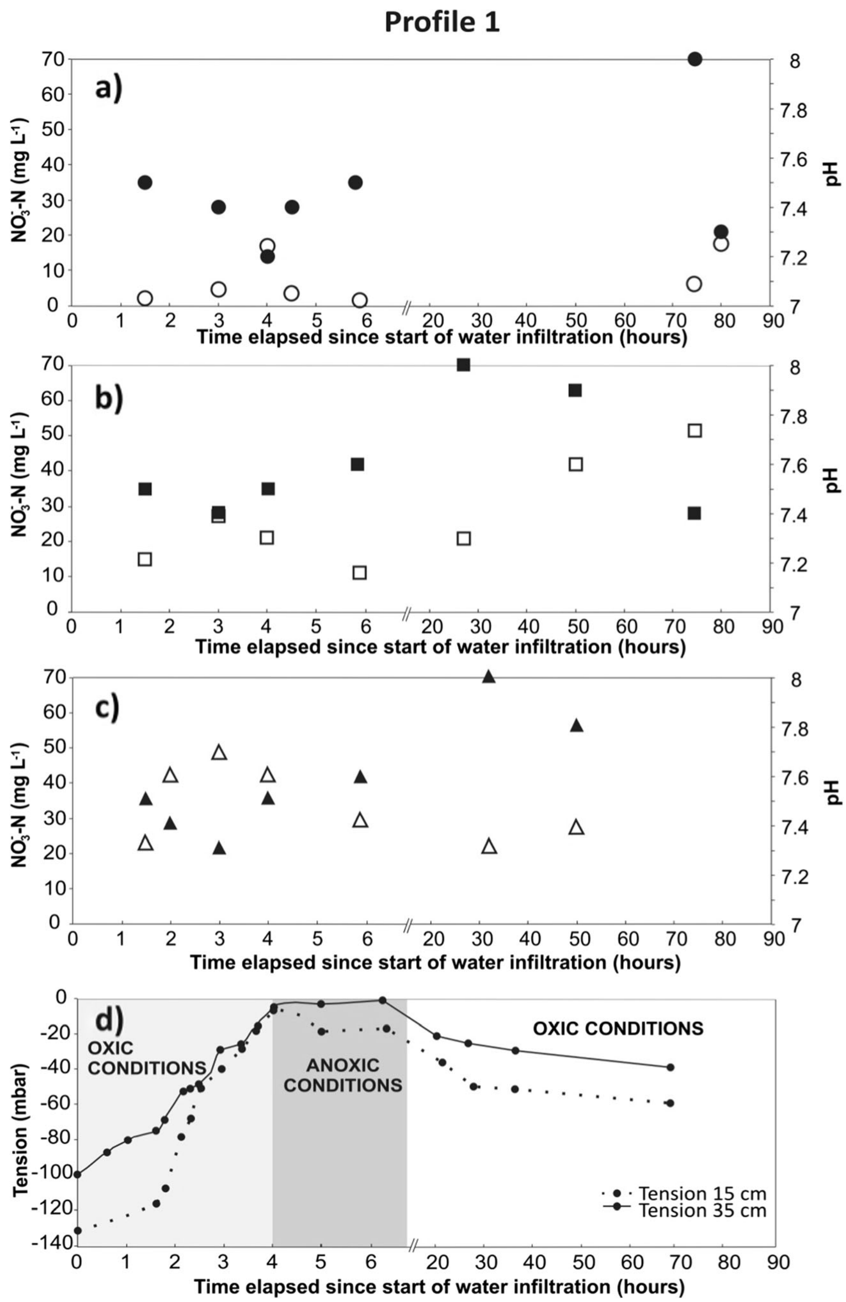
According to these results, we assume that nitrate leaching takes place particularly in the first 3 h after an irrigation event, with the rapidly percolating water through macropores. Adsorbed ammonium nitrifies after irrigation, but is retained in the soil, since vertical and lateral water movement stops. This retained nitrate is then either available for plant take-up, or denitrified by bacteria, while the soil is still moist.

### 3.4 Nitrogen balance

The field monitored in 2012 was cultivated with oat (*A. sativa* L). This field received five irrigations; extrapolating the data obtained during one of these irrigations, we estimated that 632 kg N $ha^{-1}$ was applied to the field. The Mexican Ministry for Agriculture (SAGARPA 2011) recommends fertilization doses for fodder oats of 23 kg N $ha^{-1}$ per produced ton. The crop yield was of 10,000 kg $ha^{-1}$ (dry mass), so irrigation exceeded the recommended application of 230 kg N $ha^{-1}$ by 2.7-fold. The N content in the aboveground biomass was 4.2 %; accordingly, the crop extracted 420 kg N $ha^{-1}$. This means that 65 % of the applied N was taken up by the crop, which is a larger efficiency as the 50 % commonly reported for mineral fertilizers (Grahmann et al. 2013). Fodder crops have a large vegetative growth and take up nutrients more or less constantly along the cropping cycle. Nutrient applications occur additionally in constant time intervals every 4 (oat) to 2 weeks (rye grass) and five to up to nine times during the crop cycle. Under maize, a different scenario has to be expected, since this crop needs several weeks after sawing until the vegetative growth phase starts.

To investigate N behavior of the amount of N that was not absorbed by the crop, we calculated the total N/Cl ratios in the irrigation water as well as in soil pore water and groundwater samples (Fig. 5). Chloride can be regarded as a conservative tracer since it is neither adsorbed nor transformed in the soil. Both N and Cl enter the soil with the irrigation water, and it is assumed that the total N/Cl ratio will evidence relative losses and gains of N at the different depths. A smaller total N/Cl

**Fig. 4** Concentrations of $NO_3^-$–N (mg $L^{-1}$, *open white symbols*) vs pH (*black symbols*) measured at **a** 10 cm, **b** 20 cm, and **c** 30 cm depth from the start of water infiltration till up to 90 h afterwards at profile 1 in the field monitored in 2012 and **d** water tension in mbar negative pressure measured at 15 and 35 cm depth over the same time interval




ratio would indicate a relative N loss, while a larger one would indicate a N gain.

**Table 5** Parameters of the adjusted empirical model and average percentage of $NH_4^+$-N adsorption

| Year | $1/n$ | $K_f$ | $R^2$ | % adsorbed of input ammonium |
|---|---|---|---|---|
| 2012 | 0.79 | 43 | 0.997 | 52.80 |
| 2014 | 0.78 | 36 | 0.998 | 48.75 |

$K_f$ the adsorption capacity constant, $n$ the Freundlich isotherm parameter, $R^2$ coefficient of determination

A first small N loss is observed as the irrigation water flows over the field, since the total N/Cl ratio diminishes from 0.48 to 0.40 (Fig. 5). This loss might be partly attributed to $NH_3$ volatilization, since the irrigation water and the soil have both an alkaline pH (7.7–8.4), and therefore, the equilibrium of the acid–base reaction will move towards $NH_3$ (Eq. (5)):

$$NH_4^+ + OH^- \leftrightarrow H_2O + NH_3 \quad (5)$$

$NH_4^+$–N adsorption to soil particles can be responsible also for this temporal withdrawal of N from the soil solution. The next and more important change of the total N/Cl ratio is observed within the first decimeters of the soil surface, when

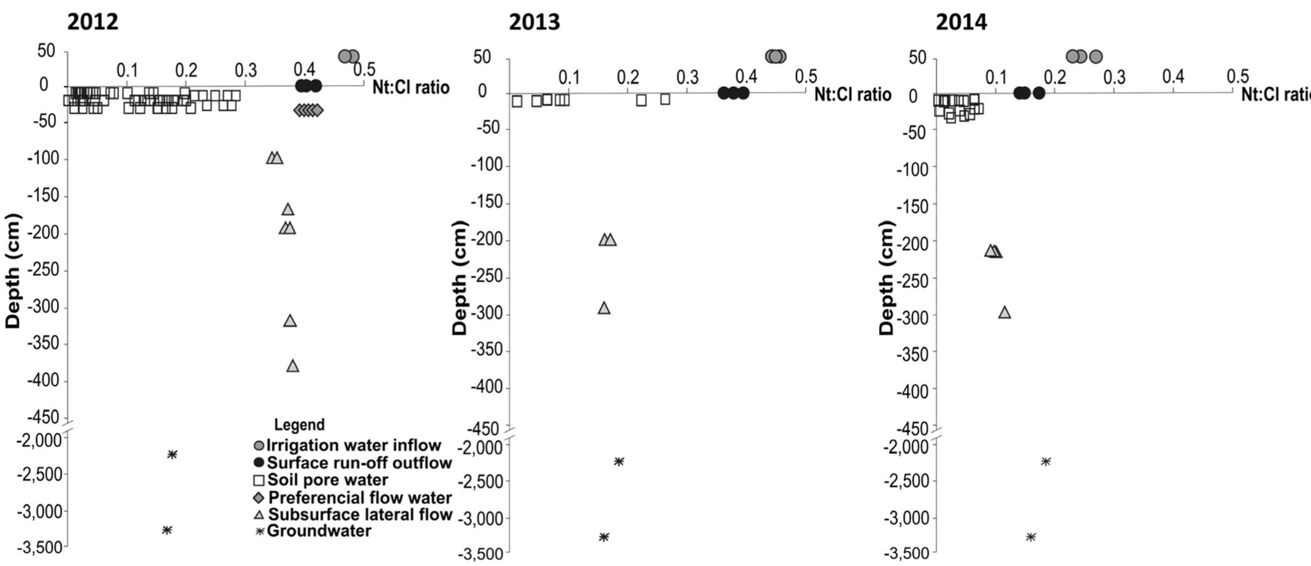



**Fig. 5** Behavior of the total N/Cl ratio in the irrigation water, soil pore water, the subsurface lateral flow, and groundwater of the different compartments within the critical zone (samples of the 2012, 2013, and 2014 monitoring campaigns)

comparing the total N/Cl ratio of the soil pore water with that of the irrigation water (Fig. 5) (0.48 to <0.3). This behavior might be caused by $NH_4^+$–N adsorption as well but also partly to losses due to denitrification of the nitrate formed by $NH_4^+$–N oxidation as the water infiltrated and saturated the soil. The preferential flow samples (rhombus in Fig. 5) do not evidence a change in the total N/Cl ratio, since water is moving relatively fast through the soil macropores, not allowing $NH_4^+$–N to oxidize, adsorb, or denitrify. The soil has two kinds of macropores: cracks and biopores. Cracks form in the smectitic clay soil due to shrinking, as the soil dries between irrigations. These cracks close as the soil swells by wetting after irrigation. Biopores are formed by roots, particularly alfalfa roots, which grow deep into the soil and form more stable pores.

The subsurface lateral flow collected in the observation wells has the same total N/Cl ratio as the surface runoff. This is again partly due to the fast infiltration by preferential flow through soil macropores, which leads to short contact times between soil and solution and therefore little or no sorption. In contrast, N in the soil solution undergoes all processes mentioned above, and therefore, the total N/Cl ratio diminishes. On the other side, a N recovery can be attributed also to nitrate leaching, i.e., to movement of nitrate containing soil pore water pressed into deeper soil layers by the infiltrating water. Subsurface lateral flow and deep percolation water beyond the rooting zone have large nitrate concentrations (40 to 42 mg $L^{-1}$ of $NO_3^-$–N). Each irrigation event produced deep percolation water (between 571 and 1953 $m^3$). Multiplication of this water volume with the determined nitrate concentration yields a N loss of 24 to 82 kg $NO_3^-$–N with each irrigation event. However, not all this nitrate reaches apparently the aquifer, which concentrations ranged between 18 and 27 mg $L^{-1}$ of $NO_3^-$–N. A decrease in nitrate concentration in the groundwater by dilution with rainwater cannot explain this large difference, since several studies have demonstrated that the aquifer recharge occurs dominantly by infiltrating irrigation water (BGS 1998), and rainwater only contributes with 13 % to the aquifer recharge (Jiménez et al. 2005). The responsible process of this smaller nitrate concentration could be denitrification, indicated by small to nil oxygen availability in the unsaturated and saturated zone, respectively, and the presence of DOC (67.7 ± 3.6 mg $L^{-1}$) as carbon source for the denitrifying bacteria. Outgassing of denitrification products is feasible, since this is an unconfined aquifer. González-Méndez et al. (2015) determined mean $N_2O$ emissions at the soil surface of nearby fields cultivated with alfalfa, maize, and rye grass of 0.06, 0.343, and 0.049 mg N $m^{-2}$ $h^{-1}$, respectively. Extrapolated over the whole crop cycle the N loss via $N_2O$ emissions would be of less than 10 to 5 kg N $ha^{-1}$ for maize and rye grass, respectively. An additional N loss due to a complete denitrification to $N_2$ is possible but difficult to verify under field conditions. Only tests with isotope marked N could give insight to the final fate of the applied N. Indirectly, the potential of complete denitrification could be analyzed by more detailed analyses of the microbial community in the vadose zone.

## 4 Conclusions

The N dynamics in this sewage effluent irrigation system consist of consecutive fast processes that occur within a few hours. When the sewage first enters the field, small gaseous ammonia losses probably occur in the running water due to the alkaline pH, and aeration that leads to nitrification as 72 to 92 % of the irrigated water infiltrates into the soil.

We learned from the monitoring campaigns that infiltrating water splits into two fluxes: a slow matrix flow infiltration and a fast preferential flow through soil macropores. Slow infiltration fills the pores saturating the soil. Organic N is probably sorbed to soil particles, but some of it is eventually also mineralized to $NH_4^+$–N, which can be adsorbed to a large extent to the soil colloids, as the batch experiments demonstrated, or oxidized to $NO_3^-$–N during the first 4 h while oxygen is still present as shown by the increasing nitrate concentrations in the soil pore water. While the soil remains saturated with water, part of the nitrate is apparently denitrified and lost to the atmosphere. Twenty-four hours after irrigation, the macropores have drained and oxygen diffuses back into the soil, promoting again nitrification.

Water infiltrating through bypass flow or preferential flow moves very fast, and its chemical composition does not change as the water constituents are unable to interact with the soil solid phase. Water that infiltrates through either kind of pores beyond the rooting zone is transported laterally at the contact between the rooting zone and the much denser tuff layer and deep percolation water. This accounts for a loss of 33 to 57 % of irrigation water and the leaching of 24 to 82 kg of $NO_3^-$–N below the root zone (>40 to 130 cm). We assume that denitrification is occurring in these deeper parts of the unsaturated zone induced by the large DOC concentrations and the relative absence of oxygen. Whether N is denitrified and lost as $N_2$ or $N_2O$ is unknown and difficult to determine. Nevertheless, this process apparently leads to an autopurification of part of the nitrate lost out of the rooting zone and helps to explain why the nitrate concentrations in the groundwater are not as high as expected.

All these processes and their relative importance have been monitored and estimated in three consecutive years in three different fields showing the general applicability of our mass balance approach and the consistency of the underlying processes. The total N/Cl ratio was used successfully to identify and explain N dynamics in the soil pore water and the groundwater and serves as a valuable indicator at least in systems with dynamic equilibrium.

## 5 Final considerations

Overall, the water efficiency of the irrigation is rather low (between 49 and 60 %) compared to other irrigation systems, as sprinkler (75 %) and drop irrigation (90 %) (Dworak et al. 2007). However, these systems require the elimination or significant reduction of suspended particles in the irrigation water to avoid clogging. Treatment of the sewage effluents would be necessary to convert to a more efficient water use system. The efficiency of N use by the crop is 65 %, which is rather large compared to conventionally fertilized agroecosystems (Grahmann et al. 2013). Since we verified N losses of up to 10 % of the applied N mainly by nitrate leaching, N applications should be diminished. Water treatment by sedimentation/flocculation (i.e., a primary waste water treatment) would reduce the N content in the irrigation water by around 10–25 % (Kivaisi 2001) and close the gap between N additions with the irrigation and crop N requirements, so that N losses could be minimized. Sprinkler systems with drop nozzles would diminish or even avoid the fast bypass flow, which is responsible for most of the nitrate leaching.

Especially interesting has been the finding that up to 25 % of the applied nitrogen and which was not taken up by the crops could possibly be denitrified to $N_2$, a process that would render a self-cleaning capacity to the system. This process deserves further investigation as does the probable temporal adsorption of $NH_4^+$ on the soil colloids.

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